

MAT303 - Probability and Random Processes

Two Marks Questions

Probability, Distributions, Hypothesis Testing, Random Processes
& Special Classes

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UNIT I: Probability and Distributions

1. Define probability and state the three axioms of probability.

Answer: Probability is a numerical measure of the likelihood of an event occurring. Axioms: (i) $0 \leq P(A) \leq 1$ for any event A , (ii) $P(S) = 1$ where S is sample space, (iii) For mutually exclusive events $A_1, A_2, \dots$: $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

2. Define conditional probability and write its formula.

Answer: Conditional probability of event A given event B has occurred is $P(A|B) = \frac{P(A \cap B)}{P(B)}$, where $P(B) > 0$. It represents the probability of A when the sample space is reduced to B .

3. State Bayes' theorem for two events.

Answer: Bayes' theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$, where $P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$. It relates forward and reverse conditional probabilities and is used for updating probabilities based on new evidence.

4. Write the general form of Bayes' theorem for partition of sample space.

Answer: For partition $\{A_1, A_2, \dots, A_n\}$ of sample space and for the event E associated with each A_i :

$$P(A_i|E) = \frac{P(E|A_i)P(A_i)}{\sum_{i=1}^n P(E|A_i)P(A_i)}$$

, where $P(A_i)$ are prior probabilities and $P(A_i|E)$ are posterior probabilities.

5. Define a random variable and distinguish between discrete and continuous random variables.

Answer: A random variable is a real-valued function defined on the sample space that assigns numerical values to outcomes of a random experiment. Discrete random variable takes countable values (e.g., die roll). Continuous random variable takes any value in an interval (e.g., voltage, temperature).

6. Define Cumulative Distribution Function (CDF) and state its properties.

Answer: CDF is $F_X(x) = P(X \leq x)$. Properties: (i) $0 \leq F_X(x) \leq 1$, (ii) $F_X(-\infty) = 0$ and $F_X(\infty) = 1$, (iii) Non-decreasing: if $x_1 < x_2$, then $F_X(x_1) \leq F_X(x_2)$, (iv) Right-continuous.

7. Define Probability Density Function (PDF) and write its relation with CDF.

Answer: PDF is the derivative of CDF: $f_X(x) = \frac{dF_X(x)}{dx}$. Conversely, $F_X(x) = \int_{-\infty}^x f_X(t)dt$. Properties: (i) $f_X(x) \geq 0$, (ii) $\int_{-\infty}^{\infty} f_X(x)dx = 1$, (iii) $P(a \leq X \leq b) = \int_a^b f_X(x)dx$.

8. Define the expected value (mean) of a random variable.

Answer: Expected value (mean) is the weighted average of all possible values. For discrete: $E(X) = \sum x_i p(x_i)$. For continuous: $E(X) = \int_{-\infty}^{\infty} x f_X(x)dx$. It represents the center of mass of the probability distribution.

9. State the properties of expectation operator.

Answer: Properties: (i) $E(c) = c$ for constant c , (ii) $E(cX) = cE(X)$, (iii) $E(X+Y) = E(X) + E(Y)$ (linearity), (iv) If X and Y are independent: $E(XY) = E(X)E(Y)$, (v) $E(g(X)) = \sum g(x_i)p(x_i)$ for discrete case.

10. Define variance and standard deviation of a random variable.

Answer: Variance measures spread: $\text{Var}(X) = E[(X-\mu)^2] = E(X^2) - [E(X)]^2$, where $\mu = E(X)$. Standard deviation is $\sigma_X = \sqrt{\text{Var}(X)}$. Properties: $\text{Var}(cX) = c^2\text{Var}(X)$, $\text{Var}(X+c) = \text{Var}(X)$.

11. Define the n -th moment about origin and n -th central moment.

Answer: n -th moment about origin: $\mu'_n = E(X^n)$. n -th central moment: $\mu_n = E[(X-\mu)^n]$ where $\mu = E(X)$. Note: $\mu_1 = 0$ (first central moment is zero), $\mu_2 = \text{Var}(X)$ (second central moment is variance).

12. Define Moment Generating Function (MGF) and state its uses.

Answer: MGF is $M_X(t) = E(e^{tX})$. For discrete: $M_X(t) = \sum e^{tx}p(x)$. For continuous: $M_X(t) = \int e^{tx}f_X(x)dx$. Uses: (i) Uniquely determines distribution, (ii) Generates moments: $\mu'_n = \left. \frac{d^n M_X(t)}{dt^n} \right|_{t=0}$, (iii) Simplifies sum of independent random variables.

13. Write the probability mass function and parameters of Binomial distribution.

Answer: Binomial distribution $B(n, p)$: $P(X = r) = {}^nC_r p^r q^{n-r}$, $r = 0, 1, 2, \dots, n$, where n is number of trials, p is success probability, $q = 1 - p$. Parameters: n (number of trials) and p (probability of success).

14. Write the mean, variance, and MGF of Binomial distribution.

Answer: For Binomial distribution $B(n, p)$: Mean = $E(X) = np$, Variance = $\text{Var}(X) = npq$ where $q = 1 - p$, MGF = $M_X(t) = (pe^t + q)^n$. Mode = $\lfloor (n+1)p \rfloor$.

15. State the probability mass function of Poisson distribution and write its mean and variance.

Answer: Poisson distribution: $P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}$, $r = 0, 1, 2, \dots$, where $\lambda > 0$ is the parameter. Mean = $E(X) = \lambda$, Variance = $\text{Var}(X) = \lambda$, MGF = $e^{\lambda(e^t - 1)}$.

16. When does Poisson distribution approximate Binomial distribution?

Answer: Poisson approximates Binomial when: (i) n is large ($n \geq 30$), (ii) p is small ($p \leq 0.05$), (iii) $np = \lambda$ is finite (typically $\lambda < 5$). Then $B(n, p) \approx \text{Poisson}(\lambda)$ where $\lambda = np$.

17. Write the probability density function of Normal distribution and its parameters.

Answer: Normal distribution $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x < \infty$. Parameters: μ (mean, location parameter) and σ^2 (variance, scale parameter). The curve is bell-shaped and symmetric about μ .

18. Write the properties of Normal distribution.

Answer: Properties: (i) Symmetric about mean μ , (ii) Mean = Median = Mode = μ , (iii) Points of inflection at $\mu \pm \sigma$, (iv) Total area = 1, (v) About 68% data within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99.7% within $\mu \pm 3\sigma$, (vi) MGF = $e^{\mu t + \frac{\sigma^2 t^2}{2}}$.

19. Define Standard Normal distribution and write the standardization formula.

Answer: Standard Normal distribution is $N(0, 1)$ with PDF $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}$. Standardization: $Z = \frac{X-\mu}{\sigma}$ transforms $X \sim N(\mu, \sigma^2)$ to $Z \sim N(0, 1)$. This allows use of standard normal tables.

20. Define Erlang distribution and write its PDF and parameters.

Answer: Erlang distribution models sum of k independent exponential random variables. PDF: $f_X(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}$, $x \geq 0$. Parameters: k (shape, positive integer) and λ (rate). Mean = $\frac{k}{\lambda}$, Variance = $\frac{k}{\lambda^2}$. Used in queuing theory.

UNIT II: Tests of Hypothesis

1. Define inferential statistics and distinguish it from descriptive statistics.

Answer: Inferential statistics uses sample data to make inferences, predictions, and generalizations about a population. Descriptive statistics summarizes and describes data features (mean, median, mode). Inferential statistics includes hypothesis testing, confidence intervals, and prediction, while descriptive focuses on data presentation.

2. What is a hypothesis in statistical testing? Distinguish between null and alternative hypotheses.

Answer: A hypothesis is a statement about a population parameter to be tested. Null hypothesis (H_0) is the statement of no effect or no difference (assumed true). Alternative hypothesis (H_1 or H_a) is the statement we want to prove (contradicts H_0).

3. State the general procedure for testing of hypothesis.

Answer: Procedure: (i) Formulate H_0 and H_1 , (ii) Choose significance level α , (iii) Select appropriate test statistic, (iv) Determine critical region, (v) Calculate test statistic from sample data, (vi) Make decision: reject or accept H_0 , (vii) Draw conclusion in context of problem.

4. Define significance level and p-value in hypothesis testing.

Answer: Significance level (α) is the probability of Type I error (rejecting true H_0), commonly set at 0.05 or 0.01. P-value is the probability of obtaining test results at least as extreme as observed, assuming H_0 is true. If $p\text{-value} < \alpha$, reject H_0 .

5. Write the test statistic for testing the significance of difference between sample proportion (p) and population proportion (P) for large samples.

Answer: The test statistic is $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$, where p is the sample proportion, P is the population proportion, $Q = 1 - P$, and n is the sample size.

6. What are the conditions required for applying large sample tests in hypothesis testing?

Answer: The conditions are: (i) Sample size $n \geq 30$, (ii) The sampling distribution approximately follows normal distribution, and (iii) For proportion tests, $nP \geq 5$ and $nQ \geq 5$.

7. State the formula for testing the significance of difference between two sample proportions p_1 and p_2 .

Answer: The test statistic is $z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$, where $P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ (pooled proportion) and $Q = 1 - P$.

8. How do you calculate the pooled proportion P when testing difference between two proportions?

Answer: The pooled proportion is calculated as $P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$, where n_1, n_2 are sample sizes and p_1, p_2 are sample proportions respectively.

9. Write the test statistic for testing significance of difference between sample mean $\bar{x}$ and population mean μ when population variance is known.

Answer: The test statistic is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$, where $\bar{x}$ is sample mean, μ is population mean, σ is population standard deviation, and n is sample size.

10. State the test statistic when population variance is unknown in large sample test for single mean.

Answer: When population variance is unknown, $z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$, where s is the sample standard deviation. For large samples, this follows approximately normal distribution.

11. Write the general test statistic for testing difference between two sample means $\bar{X}_1$ and $\bar{X}_2$.

Answer: The test statistic is $z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$, where σ_1^2, σ_2^2 are population variances and n_1, n_2 are sample sizes.

12. What is the test statistic when samples are drawn from the same population with equal variances $\sigma_1 = \sigma_2 = \sigma$?

Answer: The test statistic becomes $z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$. When σ is unknown, it is approximated by $\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$.

13. Define Student's t -distribution and state when it is used.

Answer: Student's t -distribution is a probability distribution used for small samples ($n < 30$) when population variance is unknown. It is symmetric, bell-shaped like normal but has heavier tails. As $n \rightarrow \infty$, t -distribution approaches standard normal distribution.

14. State the degrees of freedom for Student's t -distribution in one-sample test and write its properties.

Answer: Degrees of freedom $\nu = n - 1$. Properties: (i) Symmetric about zero, (ii) Mean = 0, (iii) Variance = $\frac{\nu}{\nu-2}$ for $\nu > 2$, (iv) Heavier tails than normal distribution, (v) As $\nu \rightarrow \infty$, approaches $N(0, 1)$.

15. Write the test statistic for testing difference between sample mean and population mean using Student's t -test.

Answer: The test statistic is $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ or $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}}$ with degrees of freedom $\nu = n - 1$, where $\bar{x}$ is sample mean, μ is population mean, s is sample standard deviation, and n is sample size.

16. Write the test statistic for testing difference between two sample means using Student's t -test (independent samples).

Answer: For independent samples: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$ with $\nu = n_1 + n_2 - 2$ degrees of freedom, assuming equal population variances.

17. Define F -distribution and state its applications.

Answer: F -distribution is the ratio of two independent chi-square variables divided by their degrees of freedom: $F = \frac{\chi_1^2/\nu_1}{\chi_2^2/\nu_2}$. Applications: (i) Testing equality of variances, (ii) ANOVA, (iii) Regression analysis. It is right-skewed, non-negative with parameters (ν_1, ν_2) .

18. Write the test statistic for F -test to compare two population variances and state its degrees of freedom.

Answer: $F = \frac{s_1^2}{s_2^2}$ (where $s_1^2 > s_2^2$) with degrees of freedom $(\nu_1, \nu_2) = (n_1 - 1, n_2 - 1)$, where s_1^2, s_2^2 are sample variances and n_1, n_2 are sample sizes.

19. Define Chi-square (χ^2) distribution and state its properties.

Answer: Chi-square distribution with ν degrees of freedom is the sum of squares of ν independent standard normal variables. Properties: (i) Right-skewed, (ii) Range: $[0, \infty)$, (iii) Mean = ν , (iv) Variance = 2ν , (v) As ν increases, approaches normal distribution.

20. Write the test statistic for Chi-square goodness of fit test and state its purpose.

Answer: Test statistic: $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$ with $\nu = k - 1 - m$ degrees of freedom, where O_i are observed frequencies, E_i are expected frequencies, k is number of cate-

gories, m is parameters estimated. Tests whether observed distribution fits expected distribution.

21. Explain the Chi-square test for independence of attributes and write its test statistic.

Answer: Tests whether two categorical variables are independent. Test statistic: $\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ with $\nu = (r-1)(c-1)$ degrees of freedom, where $E_{ij} = \frac{R_i \times C_j}{N}$, r is rows, c is columns, R_i is row total, C_j is column total, N is grand total.

UNIT III: Random Processes

1. Define a random process (stochastic process).

Answer: A random process is a collection of random variables indexed by time parameter t , denoted as $\{X(t), t \in T\}$. For each fixed t , $X(t)$ is a random variable. For each sample point ω , $X(t, \omega)$ is a realization or sample function of the process.

2. State the classification of random processes based on state space and time parameter.

Answer: Classification: (i) Discrete state, Discrete time (e.g., random walk), (ii) Discrete state, Continuous time (e.g., Poisson process), (iii) Continuous state, Discrete time (e.g., sampled continuous signal), (iv) Continuous state, Continuous time (e.g., thermal noise).

3. What are the methods of describing a random process?

Answer: Methods: (i) Complete probabilistic description using joint distribution functions, (ii) Mean and autocorrelation function (partial description), (iii) Sample function description, (iv) Power spectral density description. Complete description requires joint PDF of all orders.

4. Write the analytical representation of a random process.

Answer: A random process can be analytically represented as $X(t) = X_c(t) + jX_s(t)$ where $X_c(t)$ is in-phase component and $X_s(t)$ is quadrature component. For real process: $X(t) = A(t) \cos[\omega_0 t + \Theta(t)]$ where $A(t)$ is random amplitude and $\Theta(t)$ is random phase.

5. Define mean function (ensemble average) of a random process.

Answer: Mean function is $\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_X(x; t) dx$. It represents the expected value at time t averaged over all ensemble members. For discrete state: $\mu_X(t) = \sum x_i P(X(t) = x_i)$.

6. Define autocorrelation function of a random process and state its significance.

Answer: Autocorrelation function is $R_X(t_1, t_2) = E[X(t_1)X(t_2)]$. It measures the degree of correlation between process values at two different time instants t_1 and t_2 . It characterizes the time-domain behavior and frequency content of the process.

7. State the properties of autocorrelation function.

Answer: Properties: (i) $R_X(t, t) = E[X^2(t)] \geq 0$ (mean square value), (ii) $R_X(t_1, t_2) = R_X(t_2, t_1)$ (symmetric), (iii) $|R_X(t_1, t_2)| \leq \sqrt{R_X(t_1, t_1)R_X(t_2, t_2)}$, (iv) For WSS: $R_X(\tau) = R_X(-\tau)$ (even function).

8. Define cross-correlation function and state its properties.

Answer: Cross-correlation function of two processes $X(t)$ and $Y(t)$ is $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$. Properties: (i) $R_{XY}(t_1, t_2) = R_{YX}(t_2, t_1)$, (ii) $|R_{XY}(t_1, t_2)|^2 \leq R_X(t_1, t_1)R_Y(t_2, t_2)$, (iii) For WSS: $R_{XY}(\tau) = R_{YX}(-\tau)$.

9. Define autocovariance function of a random process.

Answer: Autocovariance function is $C_X(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))]$ $= R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$. It measures correlation after removing the mean. When means are zero: $C_X = R_X$.

10. Define a stationary random process and its types.

Answer: A random process is stationary if its statistical properties do not change with time. Types: (i) Strict-Sense Stationary (SSS): All joint distributions are time-invariant, (ii) Wide-Sense Stationary (WSS): Mean is constant and autocorrelation depends only on time difference.

11. State the conditions for Wide-Sense Stationary (WSS) process.

Answer: A process $X(t)$ is WSS if: (i) Mean is constant: $\mu_X(t) = \mu_X$ (independent of t), (ii) Autocorrelation depends only on time difference: $R_X(t_1, t_2) = R_X(t_2 - t_1) = R_X(\tau)$ where $\tau = t_2 - t_1$.

12. State the conditions for Strict-Sense Stationary (SSS) process.

Answer: A process is SSS if all finite-dimensional joint distributions are invariant under time shift: $F_{X(t_1), X(t_2), \dots, X(t_n)}(x_1, x_2, \dots, x_n) = F_{X(t_1+\tau), X(t_2+\tau), \dots, X(t_n+\tau)}(x_1, x_2, \dots, x_n)$ for all τ . SSS implies WSS but not vice versa.

13. Give an example of a Wide-Sense Stationary process.

Answer: Example: $X(t) = A \cos(\omega_0 t + \Theta)$ where A and ω_0 are constants, Θ is uniformly distributed over $[0, 2\pi]$. Mean: $\mu_X = 0$. Autocorrelation: $R_X(\tau) = \frac{A^2}{2} \cos(\omega_0 \tau)$ (depends only on τ). Hence WSS.

14. Define Power Spectral Density (PSD) of a random process.

Answer: Power Spectral Density $S_X(\omega)$ describes frequency content of a random process. For WSS process, it is the Fourier transform of autocorrelation: $S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-i\omega\tau} d\tau$. It represents average power per unit frequency.

15. State Wiener-Khinchin theorem.

Answer: If $X_T(\omega)$ is a F.T of truncated random process

$$X_T(t) = \begin{cases} X(t), & -T \leq t \leq T, \\ 0, & \text{elsewhere.} \end{cases}$$

where $X(t)$ is a real WSS process then the PSD is given by

$$S(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \mathbb{E} \left\{ |X_T(\omega)|^2 \right\}.$$

16. State the properties of Power Spectral Density.

Answer: Properties: (i) $S_X(\omega) \geq 0$ (non-negative), (ii) $S_X(\omega) = S_X(-\omega)$ (even function for real processes), (iii) Total average power: $R_X(0) = E[X^2(t)] = \int_{-\infty}^{\infty} S_X(\omega) df$, (iv) Units: power per unit frequency (W/Hz).

17. Define cross-power spectral density and state its relation with cross-correlation.

Answer: Cross-power spectral density of processes $X(t)$ and $Y(t)$ is $S_{XY}(\omega) = \mathcal{F}\{R_{XY}(\tau)\} = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau$. Inverse: $R_{XY}(\tau) = \int_{-\infty}^{\infty} S_{XY}(\omega) e^{i\omega\tau} df$. Generally complex-valued: $S_{XY}(\omega) = S_{YX}^*(\omega)$.

18. What is a multiplicative random process? Give an example.

Answer: A multiplicative process is formed by multiplying two random processes: $Z(t) = X(t)Y(t)$. Example: Amplitude modulation where message signal $m(t)$ multiplies carrier $\cos(\omega_c t)$: $s(t) = m(t) \cos(\omega_c t)$. Mean: $\mu_Z(t) = \mu_X(t)\mu_Y(t)$ if X and Y are independent.

19. Define a Linear Time-Invariant (LTI) system and its impulse response.

Answer: An LTI system is characterized by: (i) Linearity: $\alpha x_1(t) + \beta x_2(t) \rightarrow \alpha y_1(t) + \beta y_2(t)$, (ii) Time-invariance: $x(t - t_0) \rightarrow y(t - t_0)$. Impulse response $h(t)$ is the output when input is $\delta(t)$. System output: $Y(t) = \int_{-\infty}^{\infty} h(\tau) X(t - \tau) d\tau$.

UNIT IV: Special Classes of Random Processes

1. Define a Poisson process and state its characteristics.

Answer: Poisson process $\{X(t), t \geq 0\}$ counts the number of events occurring in $[0, t]$. Characteristics: (i) $X(0) = 0$, (ii) Independent increments, (iii) Stationary increments, (iv) $P(X(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$ where λ is rate parameter.

2. Write the mean and variance of Poisson process.

Answer: For Poisson process with rate λ : Mean = $E[X(t)] = \lambda t$, Variance = $\text{Var}[X(t)] = \lambda t$. The mean and variance are equal and proportional to time. Inter-arrival times are exponentially distributed with parameter λ .

3. State the properties of inter-arrival times in a Poisson process.

Answer: Inter-arrival times $T_1, T_2, \dots$ are: (i) Independent and identically distributed (i.i.d.), (ii) Exponentially distributed: $f_T(t) = \lambda e^{-\lambda t}$, $t \geq 0$, (iii) Memoryless property: $P(T > s + t | T > s) = P(T > t)$, (iv) Mean inter-arrival time = $\frac{1}{\lambda}$.

4. Write the autocorrelation function of a Poisson process.

Answer: The autocorrelation function (ACF) of a Poisson process $X(t)$ with rate λ is given by: $R_X(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda \min(t_1, t_2)$, where $t_1, t_2 \geq 0$. This is derived using $R_X(t_1, t_2) = E[X(t_1)X(t_2)]$ and properties of independent increments.

5. Prove that the sum of two independent Poisson processes is a Poisson process.

Answer: Let $X(t) \sim \text{Poisson}(\lambda_1)$ and $Y(t) \sim \text{Poisson}(\lambda_2)$ be independent. Then $Z(t) = X(t) + Y(t)$ has $E[Z(t)] = (\lambda_1 + \lambda_2)t$ and $E[Z^2(t)] = (\lambda_1 + \lambda_2)t + (\lambda_1 + \lambda_2)^2 t^2$. Since these match the Poisson moments, $Z(t) \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

6. State the distribution of inter-arrival times in a Poisson process and prove it.

Answer: The time interval T between successive occurrences follows exponential distribution: $f(t) = \lambda e^{-\lambda t}$, $t \geq 0$, with mean $E(T) = \frac{1}{\lambda}$. Proof: $P(T > t) = P(\text{no events in } (0, t)) = e^{-\lambda t}$. Thus CDF: $F(t) = 1 - e^{-\lambda t}$ and PDF: $f(t) = \lambda e^{-\lambda t}$.

7. Why is the difference of two independent Poisson processes not a Poisson process?

Answer: Let $X(t) \sim \text{Poisson}(\lambda_1)$, $Y(t) \sim \text{Poisson}(\lambda_2)$, and $W(t) = X(t) - Y(t)$. While $E[W(t)] = (\lambda_1 - \lambda_2)t$, the variance $\text{Var}[W(t)] = (\lambda_1 + \lambda_2)t \neq E[W(t)]$ unless

$\lambda_2 = 0$. Since Poisson requires mean equals variance, $W(t)$ is not Poisson. Also, $W(t)$ can be negative.

8. Derive the second moment $E[X^2(t)]$ for a Poisson process.

Answer: For Poisson process with rate λ : $E[X^2(t)] = \text{Var}[X(t)] + [E[X(t)]]^2 = \lambda t + (\lambda t)^2 = \lambda t + \lambda^2 t^2$. This result is used to derive the autocorrelation function and to verify that sum of Poisson processes is Poisson.

9. What is a Birth and Death process?

Answer: Birth and Death process is a continuous-time Markov chain where state changes occur in unit steps (births increase state by 1, deaths decrease by 1). Parameters: birth rate λ_n in state n , death rate μ_n in state n . Used in queuing theory and population models.

10. Write the differential-difference equations for Birth and Death process.

Answer: The forward Kolmogorov equations are: $\frac{dP_n(t)}{dt} = \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) - (\lambda_n + \mu_n)P_n(t)$ for $n > 0$, and $\frac{dP_0(t)}{dt} = \mu_1P_1(t) - \lambda_0P_0(t)$ for $n = 0$, where $P_n(t) = P(X(t) = n)$.

11. Define Pure Birth process and write its differential equation.

Answer: Pure Birth process has only births (no deaths): $\mu_n = 0$ for all n . Differential equations: $\frac{dP_n(t)}{dt} = \lambda_{n-1}P_{n-1}(t) - \lambda_nP_n(t)$ for $n \geq 1$, and $\frac{dP_0(t)}{dt} = -\lambda_0P_0(t)$ for $n = 0$. Initial condition: $P_0(0) = 1$.

12. What is Yule-Furry process and write its characteristic equation?

Answer: Yule-Furry process is a pure birth process with linear birth rate $\lambda_n = n\lambda$ (rate proportional to population size). Solution: $P_n(t) = e^{-\lambda t}(1 - e^{-\lambda t})^{n-1}$ for $n \geq 1$ with $N(0) = 1$. Mean: $E[N(t)] = e^{\lambda t}$. Models population growth.

13. Define a Renewal process and state its key property.

Answer: Renewal process $\{X(t), t \geq 0\}$ counts the number of renewals (events) by time t , where inter-arrival times $T_1, T_2, \dots$ are i.i.d. positive random variables. Key property: Times between successive renewals are independent with common distribution $F(x)$.

14. What is a renewal function? Write its definition.

Answer: Renewal function $m(t) = E[X(t)]$ is the expected number of renewals by time t . It satisfies the renewal equation: $m(t) = F(t) + \int_0^t m(t-x)dF(x)$, where $F(x)$ is the CDF of inter-arrival time. For Poisson process: $m(t) = \lambda t$.

15. State the renewal equation.

Answer: The renewal equation is $m(t) = F(t) + \int_0^t m(t-x)f(x)dx$ where $m(t)$ is the renewal function, $F(t)$ is the CDF and $f(x)$ is the PDF of inter-arrival time. This integral equation can be solved using Laplace transforms or numerical methods.

16. Define discrete state, discrete time renewal process.

Answer: In discrete state, discrete time renewal process, events occur at discrete time points $n = 0, 1, 2, \dots$ and state space is discrete. Inter-renewal times $T_1, T_2, \dots$ are i.i.d. discrete random variables. Example: Number of trials until r -th success in Bernoulli trials.

17. Define discrete state, continuous parameter renewal process.

Answer: In discrete state, continuous parameter renewal process, state space is discrete but time is continuous. The process $X(t)$ takes integer values and evolves continuously. Example: Poisson process where $X(t) \in \{0, 1, 2, \dots\}$ for continuous $t \geq 0$.

18. What is the Elementary Renewal Theorem?

Answer: Elementary Renewal Theorem states that $\lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu}$, where $m(t)$ is renewal function and $\mu = E[X]$ is mean inter-renewal time. Long-run renewal rate equals reciprocal of mean inter-arrival time. For Poisson: rate = $\lambda = \frac{1}{\mu}$.

19. State the relationship between Poisson process and exponential distribution.

Answer: In a Poisson process with rate λ : (i) Inter-arrival times are exponentially distributed: $f(t) = \lambda e^{-\lambda t}$, (ii) Time until n -th event follows Gamma (Erlang) distribution, (iii) Number of events in time t follows Poisson distribution with parameter λt .

Key Formulas Summary

Unit I - Probability and Distributions

- Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Bayes' Theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Variance: $\text{Var}(X) = E(X^2) - [E(X)]^2$
- MGF: $M_X(t) = E(e^{tX})$, $\mu'_n = \left. \frac{d^n M_X(t)}{dt^n} \right|_{t=0}$
- Binomial: $P(X = r) = {}^nC_r p^r q^{n-r}$, Mean = np , Variance = npq
- Poisson: $P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}$, Mean = Variance = λ
- Normal: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- Erlang: $f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}$, Mean = $\frac{k}{\lambda}$

Unit II - Testing of Hypothesis

- Large sample proportion: $z = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$
- Difference of proportions: $z = \frac{p_1-p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$
- Large sample mean: $z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$
- Small sample mean: $t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n-1}}}$, df = $n - 1$
- Two sample t -test: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$, df = $n_1 + n_2 - 2$
- Paired t -test: $t = \frac{\bar{d}}{\frac{s}{\sqrt{n-1}}}$, df = $n - 1$
- F -test: $F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$, df = $(n_1 - 1, n_2 - 1)$
- Chi-square: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

Unit III - Random Processes

- Mean function: $\mu_X(t) = E[X(t)]$
- Autocorrelation: $R_X(t_1, t_2) = E[X(t_1)X(t_2)]$
- Cross-correlation: $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$
- Autocovariance: $C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$
- WSS conditions: $\mu_X(t) = \mu_X$ (constant), $R_X(t_1, t_2) = R_X(\tau)$
- Power Spectral Density: $S_X(f) = \mathcal{F}\{R_X(\tau)\}$
- Wiener-Khinchin: $S_X(f) = \int_{-\infty}^{\infty} R_X(\tau)e^{-j2\pi f\tau} d\tau$
- Total power: $R_X(0) = E[X^2(t)] = \int_{-\infty}^{\infty} S_X(f)df$
- LTI system output: $Y(t) = \int_{-\infty}^{\infty} h(\tau)X(t - \tau)d\tau$

Unit IV - Special Classes of Random Processes

- Poisson process: $P(X(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$, $E[X(t)] = \lambda t$
- Inter-arrival time: $f_T(t) = \lambda e^{-\lambda t}$, $E[T] = \frac{1}{\lambda}$
- Birth-Death: $\frac{dP_n(t)}{dt} = \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) - (\lambda_n + \mu_n)P_n(t)$
- Pure Birth: $\frac{dP_n(t)}{dt} = \lambda_{n-1}P_{n-1}(t) - \lambda_n P_n(t)$
- Yule-Furry: $\lambda_n = n\lambda$, $E[X(t)] = e^{\lambda t}$
- Renewal function: $m(t) = E[X(t)] = F(t) + \int_0^t m(t-x)dF(x)$
- Elementary Renewal: $\lim_{t \rightarrow \infty} \frac{m(t)}{t} = \frac{1}{\mu}$
- Compound Poisson: $Y(t) = \sum_{i=1}^{X(t)} Z_i$, $E[Y(t)] = \lambda t E[Z]$